Research Methodology
Course Progress 44.72% (11 of 12)
MCQ and MSQ Assignment
Numerical and Short Answer Assignment
Subjective Assignment (Essay Type)
Subjective Assignment (Scan and Upload)
Programming Assignment
Week 1: Group Discussion on Research
Part 1
Part 2
Part 3
Part 4
Quiz: Week 1 : Assignment
Week 1 Feedback Form: Research Methodology
Week 1: Overview of Research
Week 2: Literature Survey, Experimental skills
Overview of Literature Survey

Assessment

Week 1 : Assignment

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-08-02, 22:52 IST.

Due date on 2026-08-05, 23:59 IST

1. What sets a PhD apart from other degrees? 1 Point

- [ ] Focus on practical skills and hands-on experience.
- [x] Development of critical thinking and analytical abilities.
- [ ] Preparation to guide others in pursuing a PhD.
- [ ] Completion of coursework and examinations.

Partially Correct. Score: 0.5 | Accepted answer is Development of critical thinking and analytical abilities., Preparation to guide others in pursuing a PhD.

2. What distinguishes a research-based degree from a typical undergraduate degree? 1 Point

- ( ) Clear and defined boundaries throughout the degree program.
- (•) Opportunities for continuous discovery and learning.
- ( ) Graduation day marking the completion of the degree.
- ( ) Acquisition of expertise and recognition in a specific field.

Yes, the answer is correct. Score: 1 | Accepted answer is Opportunities for continuous discovery and learning.

3. What is the importance of attitude and communication in the student-advisor relationship? 1 Point

- ( ) Having prior knowledge is the most important factor for a student to work towards a PhD.
- ( ) Advisors expect students to define their own research problems without any guidance.
- ( ) Students should choose advisors based on their reputation and availability.
- (•) Attitude, inclination to learn, and effective communication with the advisor are crucial.

Yes, the answer is correct. Score: 1 | Accepted answer is Attitude, inclination to learn, and effective communication with the advisor are crucial.

4. What makes research challenging for first-time researchers? 1 Point

- ( ) The need to handle routine work and drudgery.
- (•) The sudden shift in patterns from coursework to research.
- ( ) The discovery process involved in understanding the research process.
- ( ) The pressure to obtain a degree and complete the research process.

Yes, the answer is correct. Score: 1 | Accepted answer is The sudden shift in patterns from coursework to research.

5. What is one advantage of research when dealing with failure? 1 Point

- (•) Research allows for changing initial assumptions and exploring different approaches.
- ( ) Research ensures that failures are minimized and success is guaranteed.
- ( ) Research provides a predefined path to follow, for sure eliminating the possibility of failure.
- ( ) Research discourages any changes in assumptions or approaches to maintain consistency.

Yes, the answer is correct. Score: 1 | Accepted answer is Research allows for changing initial assumptions and exploring different approaches.

6. When conducting experiments and analysing results in research: 1 Point

- ( ) Each individual step will always lead to something significant.
- (•) The familiarity with the area helps in building experiments, but the results are often insignificant.
- ( ) Day-to-day experiments and results are usually significant and groundbreaking.
- ( ) Small steps are taken, and over time they contribute to something significant.

No, the answer is incorrect. Score: 0 | Accepted answer is Small steps are taken, and over time they contribute to something significant.

Read the passage below and answer the questions based on the passage.

Engineers and Scientists

We live in a world that is filled with a wide range of technologies that we use and we take for granted. The speed with which new technologies appear and disappear is constantly increasing. These days many of the gadgets we buy get outdated in a time frame of two to three years. This is a short time frame since many of the technologies in use are robust enough to physically last several years. However, in terms of features and utility, they get outdated much sooner than that. This is also a short time frame relative to how long technologies were used two or three decades ago. An example is the media used to store music. While gramophone records and cassettes lasted decades, DVDs have fast disappeared. Almost any mobile phone or computer purchased these

days become obsolete in two to three years.

Behind this vast moving world of technologies, is a worldwide army of scientists and engineers. They are constantly on the prowl for making advances to existing technologies. Many industries, companies, and educational institutions are filled with scientists and engineers. While there may be some similarities in the tasks they are involved in, there are also some differences.

An engineer in an industry setting usually faces several technical trouble shooting challenges. Such challenges, with stiff timelines, require substantial familiarity with information databases, and characterization techniques, to be successfully addressed. Addressing trouble shooting challenges are often based on the selection and use of the best available resources and substitutes. The types of questions the engineer is faced with include questions of the sort:

1. Is there a lighter material readily available which can replace the existing material in the product?
2. Why is a specific batch of the product not as good as another batch?
3. Will it be possible for this product to function with lesser parts than it currently has?
Answers to such questions enable the industry meet the demands of their customers at lower costs and with greater reliability.

A scientist in a university or laboratory setting faces a different kind of challenge. The questions the scientist wishes to answer are:
1. Why is the value of any specific property of a material what it is?
2. What is the fundamental science behind the property?
3. What are the limits, if any, to this property?
Answers to such questions enable one to design new materials and to push the capabilities of existing materials.

Based on the professional setting a materials specialist ends up in, he/she will likely get pulled into one of the above types of activities. However, having both - a good feel for the fundamental sciences, as well as an understanding of the engineering approach to relate to the real world, can greatly enhance the value of a materials scientist/engineer to his or her organization.

At the same time, it usually takes considerable experience to fully understand, appreciate, and make use of the linkages between fundamental science and the world of engineering. Experience and systematic studies provide us with the insight to make the connections between the real-world inventions and the science behind them. Such insight helps us truly appreciate the contributions of the scientist as well as the engineer in shaping our interactions with the world around us. Such insight also enables us to take our technical pursuits to greater heights and wider reaches, and hence is desirable.

7. Technologies in use today have life span of decades 1 Point

- True
- False

Yes, the answer is correct. Score: 1 | Accepted answer is False

8. Engineers aim to reduce the number of parts in a product 1 Point

- True
- False

Yes, the answer is correct. Score: 1 | Accepted answer is True

Read the passage below and answer the questions based on the passage.

Global warming refers to the idea that the temperature of the planet is, on average, increasing. It is a topic of considerable discussion and debate internationally because there are differing views on the severity of the phenomenon, the cause for the phenomenon, remedial measures to be considered and the financial impact of such remedial measures.

There is very strong experimental evidence that global warming is a real phenomenon. 10 of the hottest years recorded in history have all occurred in the last 15 years or so.

There is also strong reason to believe that the amount of carbon dioxide in the atmosphere, directly impacts the raise in temperature observed globally. This evidence can be seen in the strong correlation between temperature and atmospherical carbon dioxide data collected in the past nearly 60 years, as well as in the temperature and carbon dioxide data that can be obtained for more than 100,000 years, from ice core samples obtained from Antarctica.

Based on energy usage patterns as well as the amount of carbon dioxide released as a function of the energy usage, it is possible to see how human activity can impact carbon dioxide content in the atmosphere in a significant manner. In fact a simple calculation will show that with the present level of energy usage, assuming all of it is from crude oil, the amount of carbon dioxide in the atmosphere will double in less than 100 years.

In addition to carbon dioxide, there are other gases as well that can accelerate global warming, some much more severely than carbon dioxide. However in terms of sheer quantity in the atmosphere, carbon dioxide is the most significant contributor to global warming.

The controversy over what needs to be done to combat global warming, is purely one of short term economic impact. Nations which release the most carbon dioxide total, as well the most carbon dioxide per-capita, are both reluctant to take measures to curb the release of carbon dioxide since it impacts quality of life of their populations in the short term. However there is grave danger in assuming such a short term outlook to this issue. If this issue is not addressed urgently, the planet may become unliveable in a relatively short time and at that point, with the very existence of humanity in question, it will look completely absurd to place a few comforts above survival of the planet.

Nature controls us indirectly by causing severe weather patterns that shut down cities for days and months. It is much more prudent that we address this issue on our initiative rather than wait for nature to complete the arguments in this context.

9. Global warming is only a theoretical phenomenon 1 Point

- True
- False

Yes, the answer is correct. Score: 1 | Accepted answer is False

10. Carbon dioxide in the atmosphere contributes to global warming 1 Point

- True
- False

Yes, the answer is correct. Score: 1 | Accepted answer is True